Partially Ordered(POSET) Set and Hasse Diagram

- **Show that the relation "Congruence modulo m over the set of positive integers is an equivalence relation"**

Congruence modulo

Definition: Two integers a and b are congruent modulo m iff they have the same remainder when divided by m.

denoted by: $a \equiv b \pmod{m}$

read as: a is congruent to b modulo m.

Note:

>> $a \equiv b \pmod{m}$ means $a \bmod m = b \bmod m$.

>> $a \equiv b \pmod{m}$ iff m divides $a - b$.

Congruence modulo

Ex

$16 \equiv 13 \pmod{3}$

~ 16 mod 3 = 1
13 mod 3 = 1
also,

3 divides $16 - 13 = 3$

~ other examples

$29 \equiv 8 \pmod{7}$

$60 \equiv 0 \pmod{15}$

Proof

* Prove that Congruence modulo m is equivilance Relation.

~ Let $R = \{(a,b) \mid a \equiv b \pmod{m}\}$

1. Reflexivity

$\forall a \in A, ((a,a) \in R)$

~ $(a,a) \in R$ means

$a \equiv a \pmod{m}$ —— 1

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$\forall a \in A, ((a,a) \in R)$

- $(a,a) \in R$ means
  $a \equiv a \pmod{m}$ — (1)

- (1) is True if
  m divides $(a-a) = 0$
  0 is divisible by m so,

R is Reflexive.

2. Symmetry

$\forall a, b \in A \ ((a,b) \in R \leftrightarrow (b,a) \in R\}$

if $(a,b) \in R$ means

$a \equiv b \pmod{m}$

m divides $a-b$

so, $a-b = k \cdot m$

$$a - b = km$$

$$b - a = (-k) \cdot m$$

So, m also divides $b - a$

So, $b \equiv a \pmod{m}$

So, R is Symmetric.

3. Transitivity

$a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$

~ $a - b = k \cdot m$ — (1) $\quad$ $b - c = l \cdot m$ — (2)

~ 1 + 2

~ $a - b + b - c = km + lm$

~ $a - c = (k+l) \cdot m$

- So, m divides $a - c$. So,

$a \equiv c \pmod{m}$

- So, R is transitive.

Equivalence Classes of R

Example 1: What is the equivalence class of 2 with respect to congruence modulo 5?

Solution: The equivalence class of 2 contains all integers x such that $x \equiv 2 \pmod 5$.
i.e., x mod 5 = 2 mod 5

What is 2 mod 5?
2 mod 5 = 2

[redacted]

Example 2: What is the equivalence class of 4 with respect to congruence modulo 5?

Solution: The equivalence class of 4 contains all integers x such that $x \equiv 4 \pmod 5$.
i.e., x mod 5 = 4 mod 5

What is 4 mod 5?
4 mod 5 = 4

[redacted]

Partial Ordering Relations

- A relation R on a set A is called a partial order relation if it satisfies the following three properties:
  - Relation R is Reflexive, i.e. $aRa \ \forall \ a \in A$.
  - Relation R is Antisymmetric, i.e., aRb and $bRa \Longrightarrow a = b$.
  - Relation R is transitive, i.e., aRb and $bRc \Longrightarrow aRc$.

Partial Ordering Relations

- **Example1:** Show whether the relation $(x, y) \in R$, if, $x \geq y$ defined on the set of +ve integers is a partial order relation.

- **Example1:** Show whether the relation $(x, y) \in R$, if, $x \geq y$ defined on the set of +ve integers is a partial order relation.
- **Solution:** Consider the set A = {1, 2, 3, 4} containing four +ve integers. Find the relation for this set such as R = {(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3), (1, 1), (2, 2), (3, 3), (4, 4)}.
  - **Reflexive:** The relation is reflexive as for every $a \in A$. $(a, a) \in R$, i.e. $(1, 1), (2, 2), (3, 3), (4, 4) \in R$.
  - **Antisymmetric:** The relation is antisymmetric as whenever (a, b) and $(b, a) \in R$, we have a = b.
  - **Transitive:** The relation is transitive as whenever (a, b) and (b, c) $\in R$, we have $(a, c) \in R$.
  - **Example:** $(4, 2) \in R$ and $(2, 1) \in R$, implies $(4, 1) \in R$.
  - As the relation is reflexive, antisymmetric and transitive. Hence, it is a partial order relation.

Partial Ordering Relations

- **Example2:** Show that the relation 'Divides' defined on N is a partial order relation.

- **Solution:**
  - **Reflexive:** We have a divides a, ∀ a∈N. Therefore, relation 'Divides' is reflexive.
  - **Antisymmetric:** Let a, b, c ∈N, such that a divides b. It implies b divides a iff a = b. So, the relation is antisymmetric.
  - **Transitive:** Let a, b, c ∈N, such that a divides b and b divides c.
  - Then a divides c. Hence the relation is transitive. Thus, the relation being reflexive, antisymmetric and transitive, the relation 'divides' is a partial order relation.

Example - Assignment

- Show that the relation $R = \{(a,b) | a \subseteq b\}$ defined on the power set of set $S = \{1,2,3\}$ is a partial order relation.

POSET (Partially Ordered Set)

- Consider a relation R on a set S satisfying the following properties:
  - R is reflexive, i.e., xRx for every $x \in S$.
  - R is antisymmetric, i.e., if xRy and yRx, then $x = y$.
  - R is transitive, i.e., xRy and yRz, then xRz.
- Then R is called a partial order relation, and the set S together with partial order is called a partially order set or POSET and is denoted by $(S, \leq)$.

Comparable and Incomparable

- The elements a and b of a poset (S,R) are called comparable, if either aRb or bRa. When a and b are elements of S such that neither aRb or bRa, they are called incomparable.

Comparable and Incomparable

- **Example:** Consider A = {1, 2, 3, 5, 6, 10, 15, 30} is ordered by divisibility. Determine all the comparable and non-comparable pairs of elements of A.

- **Example:** Consider A = {1, 2, 3, 5, 6, 10, 15, 30} is ordered by divisibility. Determine all the comparable and non-comparable pairs of elements of A.
- **Solution:** The comparable pairs of elements of A are:

  {1, 2}, {1, 3}, {1, 5}, {1, 6}, {1, 10}, {1, 15}, {1, 30}
  {2, 6}, {2, 10}, {2, 30}
  {3, 6}, {3, 15}, {3, 30}
  {5, 10}, {5, 15}, {5, 30}
  {6, 30}, {10, 30}, {15, 30}

- The non-comparable pair of elements of A are:

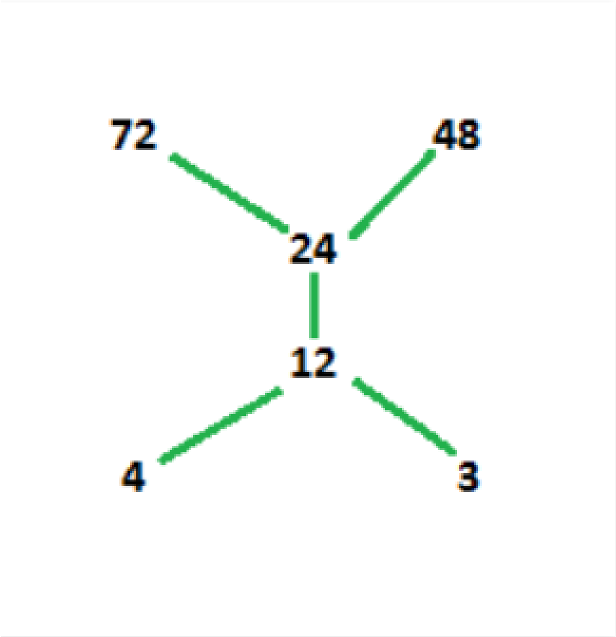
  {2, 3}, {2, 5}, {2, 15}
  {3, 5}, {3, 10}, {5, 6}, {6, 10}, {6, 15}, {10, 15}

POSET

- A **partially ordered set** (also **poset**) formalizes and generalizes the intuitive concept of an ordering, sequencing, or arrangement of the elements of a set. A poset consists of a set together with a binary relation indicating that, for certain pairs of elements in the set, one of the elements precedes the other in the ordering. The relation itself is called a "partial order."

Hasse Diagram

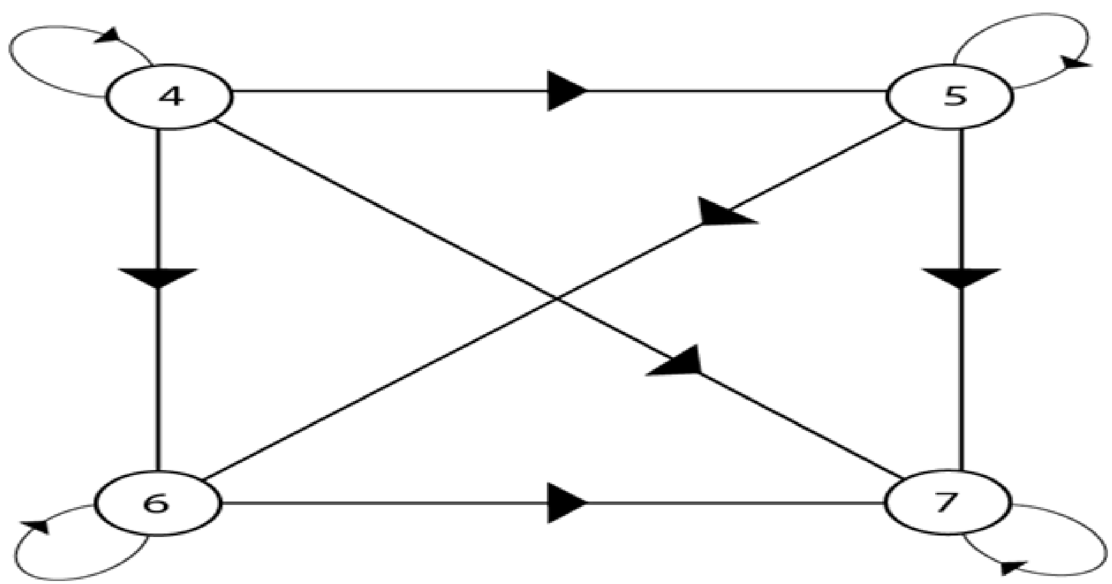
- A **Hasse diagram** is a *graphical representation* of the relation of elements of a **partially ordered set (poset)**



Hasse Diagram

- while drawing a Hasse diagram following points must be remembered :
  - The vertices in the Hasse diagram are denoted by points rather than by circles.
  - Since a partial order is reflexive, hence each vertex of A must be related to itself, so the edges from a vertex to itself are deleted in Hasse diagram.
  - Since a partial order is transitive, hence whenever aRb, bRc, we have aRc. Eliminate all edges that are implied by the transitive property in Hasse diagram, i.e., Delete edge from a to c but retain the other two edges.
  - If a vertex 'a' is connected to vertex 'b' by an edge, i.e., aRb, then the vertex 'b' appears above vertex 'a'. Therefore, the arrow may be omitted from the edges in the Hasse diagram.

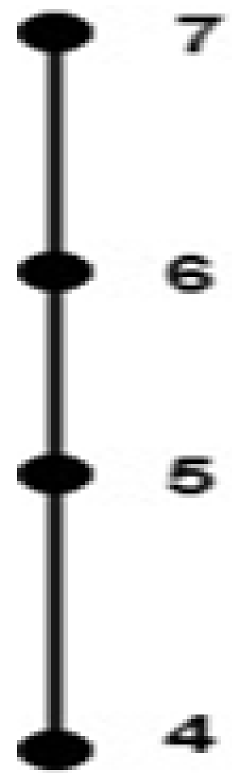
- Consider the set A = {4, 5, 6, 7}. Let R be the relation ≤ on A. Draw the directed graph and the Hasse diagram of R.



Hasse Diagram

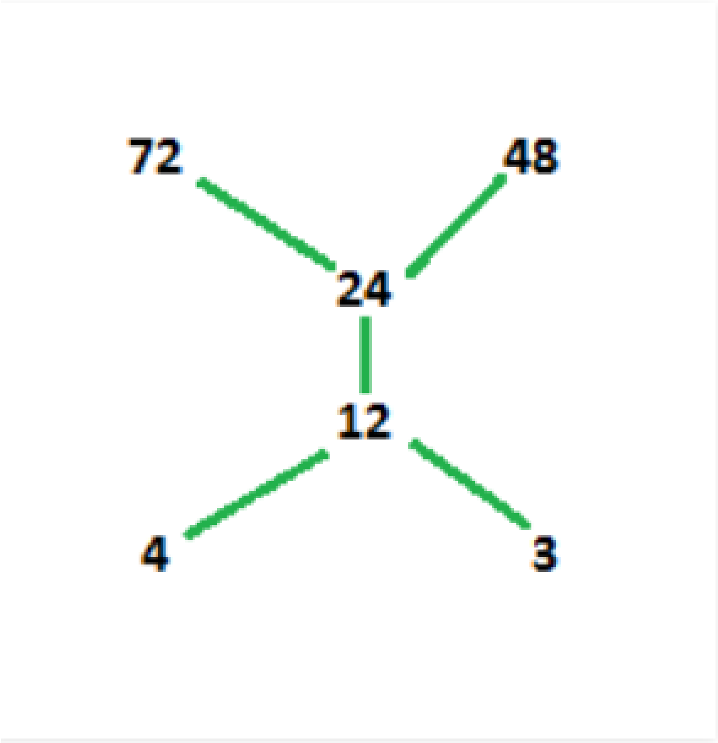
- To draw the Hasse diagram of partial order, apply the following points:
  - Delete all edges implied by reflexive property i.e. (4, 4), (5, 5), (6, 6), (7, 7)
  - Delete all edges implied by transitive property i.e. (4, 7), (5, 7), (4, 6)
  - Replace the circles representing the vertices by dots.
  - Omit the arrows.

Hasse Diagram



- **Example-1:** Draw Hasse diagram for ({3, 4, 12, 24, 48, 72}, /)
- **Explanation –** According to above given question first, we have to find the poset for the divisibility.
  Let the set is A.
  A={(3 12), (3 24), (3 48), (3 72), (4 12), (4 24), (4 48), (4 72), (12 24), (12 48), (12 72), (24 48), (24 72)}

Hasse Diagram



Hasse Diagram

- In above diagram, 3 and 4 are at same level because they are not related to each other and they are smaller than other elements in the set. The next succeeding element for 3 and 4 is 12 i.e, 12 is divisible by both 3 and 4. Then 24 is divisible by 3, 4 and 12. Hence, it is placed above 12. 24 divides both 48 and 72 but 48 does not divide 72. Hence 48 and 72 are not joined.

Hasse Diagram

Draw Hasse Diagram for following POSET

$\{(a,b) \mid a \mid b\}$

on Set $\{1, 2, 3, 4, 5, 6, 7, 8\}$

Hasse Diagram